```
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
import seaborn as sns
import tensorflow as tf
from tensorflow import keras
from tensorflow.keras import layers
import os
```

```
dataset = pd.read_csv('/content/drive/MyDrive/data/EngineFaultDB_Final.csv')
dataset.head()
```

	Fault	MAP	TPS	Force	Power	RPM	Consumption L/H	Consumption L/100KM	Speed	CO
0	0	3.549	1.889	7.428	5.227	1192.769	3.057	11.720	24.901	0.460
1	0	3.559	1.899	7.646	5.327	1145.191	2.914	11.276	25.030	0.478
2	0	3.539	1.899	7.578	5.100	1208.505	2.976	11.926	24.213	0.464
3	0	3.519	1.889	7.371	5.255	1164.127	2.908	11.786	24.407	0.471
4	0	3.549	1.879	7.405	5.177	1172.032	2.898	12.169	24.875	0.468

Next steps: Generate code with dataset New interactive sheet

```
dataset['Fault'].unique()
```

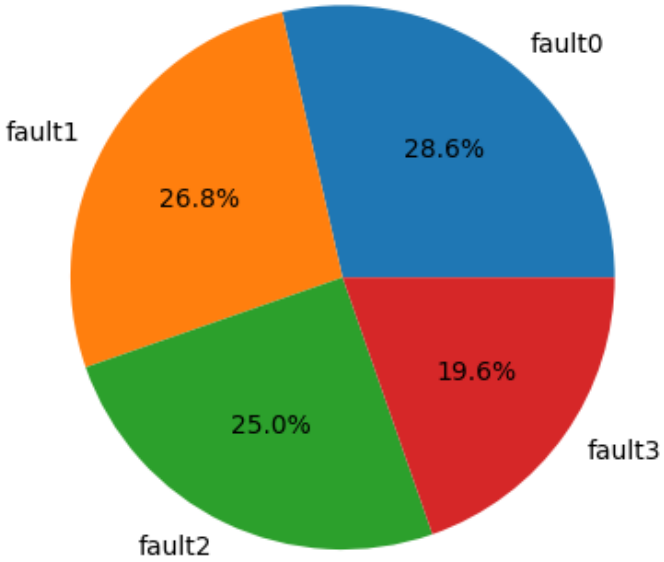
```
array([0, 1, 2, 3])
```

```
count = dataset['Fault'].value_counts()
count
```

Fault	count
0	16000
2	15000
3	14001
1	10998

dtype: int64

```
plt.pie(x = count, labels = [f'fault{i}' for i in dataset['Fault'].unique().astype(str
plt.show()
```



So first we are going to train the model on healthy data set and teach the model what healthy data looks like so when we do give it faulty data it will flag it wrong

aand for that we need a df of only healthy data letss create that

```
healthy_dataset = dataset[dataset['Fault'] == 0]
healthy_dataset.shape
```

```
(16000, 15)
```

```
healthy_dataset = healthy_dataset.drop('Fault', axis=1)
```

```
# This is the what is going to be fed into the nn it is basically converting the origi
healthy_input = healthy_dataset.values
```

```
from sklearn.model_selection import train_test_split
```

```
healthy_train, healthy_test = train_test_split(
    healthy_input, test_size=0.2, random_state=42)
print(f'Good train cound: {len(healthy_train)}' )
print(f'Good test cound: {len(healthy_test)}' )
```

```
Good train cound: 12800
Good test cound: 3200
```

```
from matplotlib.colors import Normalize
from sklearn.preprocessing import Normalizer, MinMaxScaler
from sklearn.pipeline import Pipeline

#configure out pipeline

pipeline = Pipeline([('normalize', Normalizer()),
                     ('Scalar', MinMaxScaler())])
```

```
pipeline.fit(healthy_train)
```

```
Pipeline
Pipeline(steps=[('normalize', Normalizer()), ('Scalar', MinMaxScaler())])
Normalizer
MinMaxScaler
```

```
#transform the training and validaton data with these parameters

train_transformed = pipeline.transform(healthy_train)
test_transformed = pipeline.transform(healthy_test)
```

Start coding or generate with AI.

Model Engineering

```
input_dimensions = train_transformed.shape[1]
```

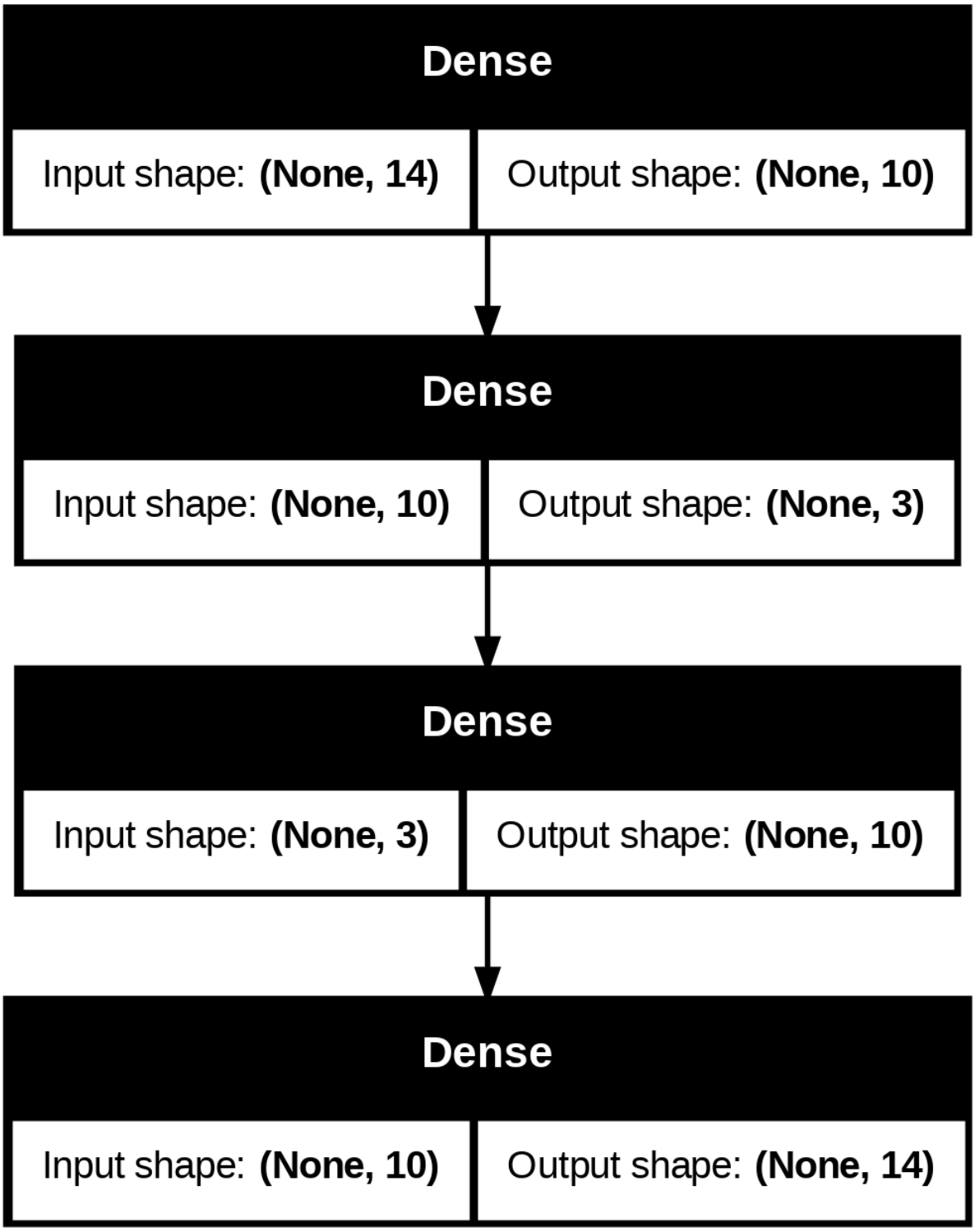
```
from sklearn import metrics
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense


model = Sequential()

model.add(Dense(10, input_dim = input_dimensions, activation = 'relu')) #inpu dimensic
model.add(Dense(3, activation = 'relu'))
model.add(Dense(10, activation = 'relu'))
model.add(Dense(input_dimensions))
```

```
/usr/local/lib/python3.12/dist-packages/keras/src/layers/core/dense.py:106: UserWarni
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

```
tf.keras.utils.plot_model(model, show_shapes=True)
```




```
model.compile(loss='mean_squared_error', optimizer = 'adam')
```

```
model.summary()
```

```
Model: "sequential"
```

Layer (type)	Output Shape	Param #
dense (Dense)	(None, 10)	150
dense_1 (Dense)	(None, 3)	33
dense_2 (Dense)	(None, 10)	40
dense_3 (Dense)	(None, 14)	154

```
Total params: 377 (1.47 KB)
Trainable params: 377 (1.47 KB)
```

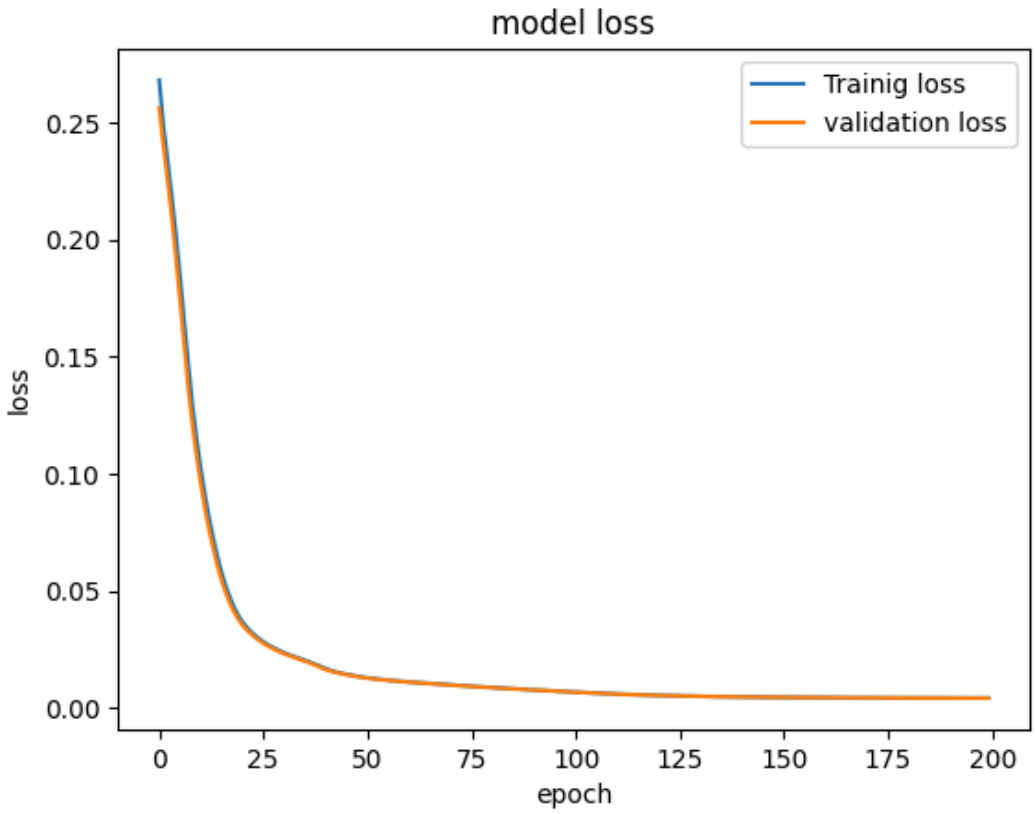
```
BATCH_SIZE =1026
EPOCHS = 200
history = model.fit(train_transformed, train_transformed,
                    verbose = 1 ,
                    epochs = EPOCHS,
                    batch_size = BATCH_SIZE,
                    validation_data = (train_transformed, train_transformed))
```

```
Epoch 1/200
13/13 ━━━━━━━━━━━━━━━━━━━━ 4s 114ms/step - loss: 0.2681 - val_loss: 0.2562
Epoch 2/200
13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - loss: 0.2489 - val_loss: 0.2403
Epoch 3/200
13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 0.2339 - val_loss: 0.2257
Epoch 4/200
13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - loss: 0.2188 - val_loss: 0.2099
Epoch 5/200
13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - loss: 0.2025 - val_loss: 0.1927
Epoch 6/200
13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 0.1843 - val_loss: 0.1736
Epoch 7/200
13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - loss: 0.1652 - val_loss: 0.1546
Epoch 8/200
13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 0.1464 - val_loss: 0.1363
Epoch 9/200
13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - loss: 0.1291 - val_loss: 0.1206
Epoch 10/200
13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - loss: 0.1145 - val_loss: 0.1072
Epoch 11/200
13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - loss: 0.1019 - val_loss: 0.0955
Epoch 12/200
13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - loss: 0.0908 - val_loss: 0.0849
Epoch 13/200
13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - loss: 0.0808 - val_loss: 0.0757
Epoch 14/200
13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - loss: 0.0722 - val_loss: 0.0678
Epoch 15/200
13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 11ms/step - loss: 0.0644 - val_loss: 0.0604
Epoch 16/200
13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - loss: 0.0576 - val_loss: 0.0543
Epoch 17/200
13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - loss: 0.0519 - val_loss: 0.0491
Epoch 18/200
13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - loss: 0.0470 - val_loss: 0.0445
Epoch 19/200
13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 11ms/step - loss: 0.0428 - val_loss: 0.0408
Epoch 20/200
13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - loss: 0.0394 - val_loss: 0.0378
```

```
Epoch 21/200
13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - loss: 0.0366 - val_loss: 0.0353
Epoch 22/200
13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - loss: 0.0344 - val_loss: 0.0333
Epoch 23/200
13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - loss: 0.0325 - val_loss: 0.0316
Epoch 24/200
13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - loss: 0.0309 - val_loss: 0.0301
Epoch 25/200
13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - loss: 0.0295 - val_loss: 0.0287
Epoch 26/200
13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 11ms/step - loss: 0.0282 - val_loss: 0.0275
Epoch 27/200
13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - loss: 0.0270 - val_loss: 0.0264
Epoch 28/200
13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - loss: 0.0260 - val_loss: 0.0255
Epoch 29/200
```

```
plt.plot(history.history['loss'], label = 'Trainig loss')
plt.plot(history.history['val_loss'], label = 'validation loss')
plt.title('model loss')
plt.ylabel('loss')
plt.xlabel('epoch')
plt.legend()
```

```
<matplotlib.legend.Legend at 0x7a46bd744080>
```



Start coding or generate with AI.

now our model knowas what good data looks like lets feed it data with anomalies

```
fault1_dataset = dataset[dataset['Fault'] == 1]
fault1_dataset.shape

fault1_dataset = fault1_dataset.drop('Fault', axis=1)
```

```
fault1_input = fault1_dataset.values
faulty_transformeddata = pipeline.transform(fault1_input)
```

```
reconstruction_faulty  = model.predict(faulty_transformeddata)

reconstructions_error_normal_train = np.mean(np.abs(faulty_transformeddata  - recons
```

```
344/344 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step
```

```
reconstruction_faulty  = model.predict(faulty_transformeddata)

reconstructions_error_normal_train = np.mean(np.abs(train_transformed  - reconstruct
```

```
344/344 ━━━━━━━━━━━━━━━━━━━━ 1s 1ms/step
---------------------------------------------------------------------------
ValueError                                Traceback (most recent call last)
/tmp/ipykernel_4257/2859463692.py in <cell line: 0>()
      1 reconstruction_faulty  = model.predict(faulty_transformeddata)
      2
----> 3 reconstructions_error_normal_train = np.mean(np.abs(train_transformed  -
reconstruction_faulty), axis = 1)

ValueError: operands could not be broadcast together with shapes (12800,14)
(10998,14)
```

Next steps: Explain error